Exact Gaussian energy and posterior identification of AND for ReLU networks

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Abstract

We determine the exact minimum Euclidean variation norm of a shallow ReLU score that represents AND on the full Boolean cube with signed output margin at least 1/2. For odd $q \geq 3$ the value is $(2q-1)/(2\sqrt{q+1})$; we give a separate closed formula for even q and the two small dimensions. The corresponding minimum under the isotropic Gaussian weight energy defined here is attained at every width at least $1 + \binom{q}{2}$ for $q \geq 3$. A weighted two-design supplies this polynomial width bound. Equality in an explicit dual certificate also identifies every globally margin-valid energy minimizer after only the positive all-plus input and its q negative neighbors: all have the AND sign function. Consequently, at each such fixed width, the actual globally conditioned Gaussian posterior concentrates on AND in the strong-prior limit after these $q+1$ labels. The target has a shallow teacher of width $q+1$ with all weights at most one. These results do not bound the number of online mistakes on arbitrary input orders.

1 Model and results

The global output-margin condition makes minimum-energy prediction a problem over all unobserved Boolean labels as well as the observed ones. A formula for the energy of a target alone therefore does not identify the posterior after a training sample. We give both an exact energy formula for AND and a sample on which every globally admissible energy minimizer has the target's full sign function. The latter statement concerns the whole posterior, including tied parameter minimizers. It is not a mistake bound for arbitrary input orders.

For $q \geq 1$, let $f_q : \{-1,1\}^q \to \{-1,1\}$ be positive only at $x^0 = (1,\dots,1)$. Write $\bar{x} = (1,x)$ and consider

$$F_{W,v}(x) = \sum_{j=1}^{N} v_j \,\mathrm{ReLU}(W_j \bar{x}), \qquad \mathrm{ReLU}(t) = \max\{t,0\}. \tag{1}$$

The constant input is the only source of bias. There is one hidden ReLU layer and a linear readout. The independent Gaussian weights are

$$W_{ji} \sim N(0, N\sigma^2/q), \qquad v_j \sim N(0,\sigma^2/N). \tag{2}$$

Their density is proportional to $\exp[-K_N/(2\sigma^2)]$, where

$$K_N(W,v) = \frac{q}{N}\|W\|_F^2 + N\|v\|_2^2. \tag{3}$$

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Before observing labels, condition on

$$|F_{W,v}(x)| > \frac{1}{2} \quad \text{for every } x \in \{-1,1\}^q. \tag{4}$$

Then condition on exact observed signs. All strong-prior limits below mean $\sigma \downarrow 0$ at a fixed finite q, N.

For a real-valued function F on the cube, define its Euclidean ReLU variation norm by

$$\|F\|_{\mathcal{V}} = \inf\left\{ \sum_j |c_j| \|u_j\|_2 : F(x) = \sum_j c_j \,\mathrm{ReLU}(u_j \bar{x}) \text{ on the cube} \right\},$$

where the infimum ranges over finite representations. Let

$$V_q = \inf\left\{ \|F\|_{\mathcal{V}} : f_q(x)F(x) \geq \frac{1}{2} \text{ on the cube} \right\}.$$

Theorem 1 (Exact norm and Gaussian energy)**.** *The exact values are*

$$V_q = \begin{cases} 1/\sqrt{2}, & q = 1, \\ (\sqrt{2}+\sqrt{6})/4, & q = 2, \\ (2q-1)/(2\sqrt{q+1}), & q \geq 3 \text{ odd}, \\ \dfrac{2q^2 + \sqrt{q(q-2)}}{2\sqrt{q(q+1)(q+2)}}, & q \geq 4 \text{ even}. \end{cases} \tag{5}$$

Put $N_q = 2$ for $q \leq 2$, and $N_q = 1 + \binom{q}{2}$ for $q \geq 3$. For every integer $N \geq N_q$, the minimum of (3) subject to $f_q(x)F_{W,v}(x) \geq 1/2$ on the full cube is exactly

$$2\sqrt{q}\, V_q.$$

The strict-margin infimum has the same value. The norm lower bound holds for every width; the stated width threshold is sufficient and is not asserted necessary.

For $1 \leq i \leq q$, let $x^i = x^0 - 2e_i$. Call the $q+1$ examples $(x^0, +1), (x^1, -1), \ldots, (x^q, -1)$ the *star sample*.

Theorem 2 (Identification from the star sample)**.** *Fix $N \geq N_q$. Every minimum-energy network satisfying the star sample and the global weak absolute-margin condition has the full sign function f_q. Under (2)–(4), after conditioning on the star sample,*

$$\mathbb{P}_\sigma(\operatorname{sign} F_{W,v} = f_q \text{ on the full cube} \mid \text{star sample}) \longrightarrow 1.$$

The probability remains at least this large after any further target-consistent labels. In particular, at fixed q, N, the expected number of subsequent sampling mistakes tends to zero on any continuation without repetitions.

Proposition 3 (A bounded teacher)**.** *For $q \geq 3$, f_q has a depth-two teacher of width $q+1$ with every individual weight at most one and strict global output margin $q/[2(q-1)] > 1/2$. For $q = 1, 2$, width two suffices with the same weight bound. Thus the width thresholds in Theorems 1 and 2 are above these explicit bounded teacher widths. A bounded-fan-in Boolean circuit for f_q uses $q-1$ AND gates when $q \geq 2$.*

2 A dual certificate for all ReLU atoms

We first treat $q \geq 3$. Define

$$\begin{gathered} n = q+1, \qquad k = \lceil n/2 \rceil, \qquad \delta = 2k - n \in \{0,1\}, \\ R = \sqrt{(q-1)^2 - \delta^2}, \qquad \varepsilon = \frac{q-1-R}{n}, \qquad A = q-1-\varepsilon, \\ D = 2\sqrt{\frac{k(n-k)}{n}} = \sqrt{n - \frac{\delta^2}{n}}. \end{gathered} \tag{6}$$

Here $R > 0$, $A > 0$, and $0 \leq \varepsilon < 1$. For a score H, put

$$T_A(H) = AH(x^0) - \sum_{i=1}^{q} H(x^i). \tag{7}$$

Lemma 4 (Atom bound and equality)**.** *For every $u = (b, w) \in \mathbb{R}^{q+1}$,*

$$|T_A(\mathrm{ReLU}(u\bar{x}))| \leq D\|u\|_2. \tag{8}$$

Among unit atom vectors, positive equality holds only at

$$H_+(x) = \frac{1}{D}\,\mathrm{ReLU}\left(-(1+\varepsilon) + (1-\varepsilon)\sum_i x_i\right). \tag{9}$$

Negative equality atoms all vanish at x^0 and have the form

$$H_S^{(j)}(x) = \mathrm{ReLU}\left(\frac{2}{D}\left[\frac{j}{n}\left(1 + \sum_i x_i\right) - \sum_{i \in S} x_i\right]\right), \quad |S| = j, \tag{10}$$

where j maximizes $j(n-j)$ among $0 \leq j \leq q$. Thus $j = k$ for odd q; for even q, either $j = q/2$ or $j = q/2 + 1$.

Proof. Write $s = b + \sum_i w_i$. If $s < 0$, then $T_A \leq 0$. If $s \geq 0$, using $\mathrm{ReLU}(t) \geq t$ gives

$$T_A \leq (A - q + 2)s - 2b = (-1-\varepsilon)b + (1-\varepsilon)\sum_i w_i.$$

The coefficient vector u_+ in this last expression satisfies

$$\|u_+\|_2^2 = n - 2(q-1)\varepsilon + n\varepsilon^2 = n - \delta^2/n = D^2.$$

Cauchy–Schwarz proves the positive bound.

For the negative bound, expand a sum of ReLUs as a maximum over subsets $S \subseteq [q]$, and write $j = |S|$. Let $e = (1, \ldots, 1) \in \mathbb{R}^n$, and let $a_S \in \mathbb{R}^n$ be the indicator of the selected input positions, zero at the constant position. On the halfspace $s \leq 0$, the linear coefficient in $-T_A$ is $je - 2a_S$, whose inner product with e is $j(q-1) \geq 0$. On $s \geq 0$, the coefficient is $(j - A)e - 2a_S$, and

$$\langle (j-A)e - 2a_S, e\rangle = j(q-1) - An \leq q(q-1) - An = -R < 0.$$

In either case, the component along e contributes a nonpositive inner product with the allowed atom vector. The perpendicular projection of the coefficient is

$$-2(a_S - je/n), \qquad \|-2(a_S - je/n)\|_2^2 = \frac{4j(n-j)}{n} \leq D^2.$$

This proves the negative bound on every activation cone.

Positive equality forces equality in Cauchy–Schwarz, hence the unit vector u_+/D. Its center preactivation is R/D, and its neighbor preactivations are $[R-2(1-\varepsilon)]/D \geq 0$. The last inequality follows directly for $q = 3, 4$; for $q \geq 5$ use $R \geq q-2 \geq 3$. Thus it is indeed an equality atom.

For negative equality, the projection norm forces $j(n-j)$ to be maximal. Such j is nonzero, so both halfspace coefficients have a strictly adverse component along e. Equality therefore requires $s = 0$, followed by equality in the perpendicular Cauchy–Schwarz step. This gives exactly (10). Conversely these vectors have unit norm, vanish at x^0, and attain $-D$. □

Any score fitting the star margins satisfies

$$T_A(F) \geq \frac{A+q}{2}, \qquad \|F\|_{\mathcal{V}} \geq \frac{A+q}{2D}. \tag{11}$$

This is a bound on every nonlinear competitor, not only a symmetric or width-one representation.

3 A full-cube witness with polynomially many atoms

We use the negative atoms (10) with the upper size $j = k = \lceil n/2 \rceil$. Choose a distribution on these k-subsets whose first and pair moments are

$$\mathbb{P}(i \in S) = \frac{k}{q}, \qquad \mathbb{P}(i, j \in S) = \frac{k(k-1)}{q(q-1)} \quad (i \neq j). \tag{12}$$

The uniform distribution is one choice. Put $H_- = \mathbb{E}_S H_S^{(k)}$.

Lemma 5 (The negative atom average)**.** *At every cube point other than x^0,*

$$H_-(x) \geq D/q.$$

Equality holds at each one-coordinate neighbor of x^0.

Proof. Suppose $h \geq 1$ coordinates of x are negative. Let J count how many of those positions belong to S, and put $Z = nJ - kh$. Then

$$H_-(x) = \frac{4}{nD}\mathbb{E}Z_+.$$

The random variable Z is the sum of h population entries, each equal to $n-k$ at a selected coordinate and $-k$ otherwise. Consequently $-kh \leq Z \leq (n-k)h$, so pointwise

$$Z^2 \leq h[(n-k)Z_+ + kZ_-] = h[nZ_+ - kZ].$$

The moments (12) give $\mathbb{E}Z = kh/q$ and

$$\frac{\mathbb{E}Z^2}{nh} + \frac{k}{n}\mathbb{E}Z = \frac{k(n-k)}{q} + \frac{(h-1)k(2k-n)}{q(q-1)} \geq \frac{k(n-k)}{q}. \tag{13}$$

For completeness, a population entry $Y_i = n1_{i\in S} - k$ has

$$\mathbb{E}Y_i^2 = \frac{k[n^2-(q+2)k]}{q}, \qquad \mathbb{E}Y_iY_j = \frac{k[(q+3)k-n^2]}{q(q-1)} \quad (i \neq j),$$

which yield (13) by expanding the square of the sum. Taking expectations in the pointwise bound proves $\mathbb{E}Z_+ \geq k(n-k)/q$. Since $D^2 = 4k(n-k)/n$, the claimed lower bound follows. At $h = 1$, the direct two-value calculation is an equality. □

Set

$$\alpha = \frac{D}{2R}, \qquad \beta = \frac{q}{D}\left(1 - \frac{1-\varepsilon}{R}\right), \qquad F = \alpha H_+ - \beta H_-. \tag{14}$$

Both coefficients are positive. At the center, $H_- = 0$ and $H_+ = R/D$, so $F(x^0) = 1/2$. At any other point, H_+ is no larger than its neighbor value. Lemma 5 therefore gives

$$F(x) \le \frac{\alpha}{D}[R - 2(1-\varepsilon)] - \frac{\beta D}{q} = -\frac{1}{2}.$$

The variation cost matches (11):

$$\alpha + \beta = \frac{A+q}{2D}. \tag{15}$$

One way to verify the identity is to use $D^2 - 2q(1-\varepsilon) = -(1+\varepsilon)R$. Substituting (6) into (15) gives the odd and even formulas in (5).

Lemma 6 (Quadratic support size)**.** *A distribution satisfying (12) can be supported on at most $\binom{q}{2}$ subsets. Hence (14) can use at most $1 + \binom{q}{2}$ ReLU atoms.*

Proof. Let $M = \binom{q}{2}$, and assign to each k-subset its pair-indicator vector $v_S = (1_{i,j\in S})_{i<j} \in \mathbb{R}^M$. Every such vector lies in the affine hyperplane $\sum_{i<j}(v_S)_{ij} = \binom{k}{2}$, of dimension at most $M-1$. The uniform pair-moment vector is in their convex hull, so Carathéodory's theorem represents it using at most M vectors. Equivalently, start with the uniform distribution and successively eliminate an affine dependence whenever its support exceeds M.

Matching pair moments also matches first moments because $k \ge 2$ and

$$\sum_{j\neq i} 1_{i,j\in S} = (k-1)1_{i\in S}.$$

This proves (12) with the claimed support. □

Remark 7 (Hadamard improvement)**.** *When q is odd and a Hadamard matrix of order $q+1$ exists, normalize its first row and column to be positive. Its q remaining rows have zero sum. Their negative input positions form $k = (q+1)/2$ subsets; column orthogonality gives (12). Thus q negative atoms and one positive atom suffice, reducing the width threshold to $q+1$. No general Hadamard existence assumption is used in the quadratic width bound.*

4 Small dimensions

For $q = 2$, use $A = 2/\sqrt{3}$ and $D = \sqrt{8/3}$ in (7). The negative halfspace-projection argument still applies: the perpendicular norm is at most D, and $j(q-1) - A(q+1) = j - 2\sqrt{3} < 0$ for $j \le 2$. For the positive bound, convexity gives

$$H(x^1) + H(x^2) \ge 2\,\mathrm{ReLU}(b).$$

When the center preactivation is nonnegative,

$$T_A(H) \le A(b + w_1 + w_2) - 2\,\mathrm{ReLU}(b) \le A(w_1 + w_2) \le D\|(b, w)\|_2,$$

because $0 < A < 2$. A negative center preactivation gives $T_A \le 0$. Equality on the positive side holds only at $H_+(x) = \mathrm{ReLU}(x_1 + x_2)/\sqrt{2}$. Negative equality atoms vanish at the center, by the same strict halfspace argument.

The two-atom witness

$$F(x) = \frac{1}{4}\big[\mathrm{ReLU}(x_1 + x_2) - \mathrm{ReLU}(2 - x_1 - x_2)\big]$$

has outputs $1/2, -1/2, -1$ on the three Hamming layers. Its cost is $(\sqrt{2} + \sqrt{6})/4$, equal to the dual lower bound $(A + 2)/(2D)$.

For $q = 1$, the atom difference $H(1) - H(-1)$ is at most $\sqrt{2}\|(b, w)\|_2$ in absolute value. If both preactivations are nonnegative, use $|b| \geq |w|$ to bound $2|w|$; if exactly one is nonnegative, use Cauchy–Schwarz on $b \pm w$; if neither is nonnegative, the difference vanishes. The two required margins have difference at least one. The matching witness is

$$F(x) = \frac{1}{4}\big[\mathrm{ReLU}(1 + x) - \mathrm{ReLU}(1 - x)\big] = x/2 \quad \text{on } \{-1, 1\}.$$

Its variation cost is $1/\sqrt{2}$.

5 Gaussian energy and posterior identification

Proof of Theorem 1. The preceding dual bounds and witnesses prove all norm formulas. For each neuron, arithmetic–geometric mean gives

$$\frac{q}{N}\|W_j\|_2^2 + N v_j^2 \geq 2\sqrt{q}\,|v_j|\|W_j\|_2.$$

Summing proves the Gaussian lower bound. Conversely, for a unit atom with coefficient $c \neq 0$, choose a positive row scale t with $t^2 = N|c|/\sqrt{q}$, and use readout coefficient c/t. Positive homogeneity preserves the atom's contribution, and its energy is exactly $2\sqrt{q}\,|c|$. Apply this to the witnesses and pad unused neurons by zeros. Lemma 6 and the two-atom small-dimensional witnesses give the stated thresholds. Finally, readout amplification by factors tending down to one proves equality of strict and weak infima. □

Lemma 8 (Sign rigidity after the star sample)**.** *For $N \geq N_q$, every minimum-energy network fitting the star sample and global weak absolute margins has sign function f_q.*

Proof. For $q \geq 3$, the full-cube witness and the star dual show that the minimum energy is $2\sqrt{q}\,V_q$. Equality must hold throughout

$$K_N \geq 2\sqrt{q}\sum_j |v_j|\|W_j\|_2 \geq \frac{2\sqrt{q}}{D} T_A(F) \geq 2\sqrt{q}\,V_q.$$

Every nonzero atom therefore attains dual equality, with its coefficient sign agreeing with its dual sign. By Lemma 4, the score has the form

$$F = \alpha' H_+ - \sum_S \beta_S H_S, \qquad \alpha', \beta_S \geq 0,$$

where each negative atom is nonnegative and vanishes at x^0. Equality in the star-margin bound forces all star margins to equal $1/2$, in particular $F(x^0) = 1/2$. Thus $\alpha' = D/(2R)$. At every other point, the strictly smaller value of H_+ gives

$$F(x) \leq \alpha' H_+(x) < \alpha' H_+(x^0) = 1/2.$$

Global weak absolute margin forces $F(x) \leq -1/2$.

For $q = 2$, use the positive and negative equality cases from the small-dimensional argument. The positive equality atom is zero at every noncenter cube point, so the same conclusion holds. For $q = 1$, the star sample is the full domain. □

Proof of Theorem 2. Let $\mathcal{E}$ be the event of strict global absolute margin and the star labels. Its energy infimum is $E_* = 2\sqrt{q}\,V_q$, by the witnesses and readout amplification. Let $\mathcal{B}$ be the closed set satisfying global weak absolute margins and the star labels but assigning a positive label to some other cube point. This is a finite union of closed sign cells. If nonempty, it attains an energy minimum by coercivity. Lemma 8 implies

$$\inf_{\mathcal{B}} K_N > E_*.$$

For completeness, this strict energy gap controls the actual finite-dimensional Gaussian posterior. Choose a strict feasible point of energy below $\inf_{\mathcal{B}} K_N$, and a sufficiently small ball around it contained in $\mathcal{E}$. Integration of the Gaussian density over this ball gives a lower bound on $\mathbb{P}_\sigma(\mathcal{E})$ with a smaller exponential rate than the Gaussian upper bound for the closed set $\mathcal{B}$. Thus

$$\mathbb{P}_\sigma(\mathcal{B} \mid \mathcal{E}) \longrightarrow 0.$$

The statement is trivial if $\mathcal{B}$ is empty. This proves concentration on the full AND function without selecting any particular parameter minimizer.

Additional target-consistent labels preserve the entire AND function cell and can only remove other cells. Its posterior probability therefore cannot decrease. A continuation without repetitions has at most $2^q - q - 1$ remaining inputs, so its expected number of sampling mistakes is at most this number times the post-star probability of a non-AND function, which tends to zero. □

6 Teacher and scope

Proof of Proposition 3. For $q \geq 3$, use q identical hidden neurons

$$\mathrm{ReLU}\left(\frac{\sum_i x_i - (q-1)}{q-1}\right),$$

each with readout coefficient one. Each is $1/(q-1)$ at the center and zero elsewhere. Add a constant neuron ReLU(1) with readout coefficient $-q/[2(q-1)]$. The output is exactly $q/[2(q-1)]$ at the center and its negative elsewhere. All individual weights have magnitude at most one.

For $q = 2$, the width-two score

$$\frac{1}{2}\,\mathrm{ReLU}(x_1 + x_2) - \mathrm{ReLU}\left(1 - \frac{x_1 + x_2}{2}\right)$$

has outputs $1, -1, -2$ on the three Hamming layers and all weights at most one. For $q = 1$, use $[\mathrm{ReLU}(1 + x) - \mathrm{ReLU}(1 - x)]/2 = x$ on the cube. The circuit statement is the usual binary AND tree. □

The norm values are intrinsic to ordinary ReLU and the isotropic Gaussian energy defined here. Their noninteger form does not by itself supply an ordering of exponentially many competing functions. Nor does the displayed construction prove an additive norm formula for multiple gadgets: an AND of disjoint AND blocks is simply one AND. The posterior theorem applies once the star labels have been observed. It leaves open how many mistakes can occur before then on an arbitrary permutation of the cube.